

Giacomo Saccaggi

SENIOR AI / ML ENGINEER

Senior AI/ML Engineer and Technical Owner with 7+ years of experience building scalable, production-ready AI systems. A proactive builder who has developed end-to-end applications since university, bridging the gap between hardware (IoT), software engineering (Full-stack), and statistical theory. Proven track record of high-stakes leadership, including saving €800,000/year in infrastructure costs and protecting 25% of corporate digital revenue through strategic architectural pivots. Expert in MLOps, Privacy-Preserving Analytics, and Multi-Agent Orchestration.

CONTACT

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 [giacomosaccaggi.github.io](https://github.com/giacomosaccaggi)

LANGUAGES

Italian:	Native
English:	Proficiency
Spanish:	Intermediate
German:	Beginner (Currently learning)

SKILLS

Core Programming: Python (Expert), SQL (Expert), R (Expert), Kotlin/Java (JetBrains SDK), Scala, VB.net.

MLOps & Cloud Infrastructure: Azure (Certified: AI/ML/Data), AWS (EC2, S3, RDS, SageMaker), GCP (Compute Engine), Docker, Kubernetes, Airflow, GitHub Actions (CI/CD), IBM Watson, .

Data Engineering & Search: Polars, Spark AQE, ElasticSearch, DuckDB, High-throughput Data Pipelines, JDBC, Presto/Trino, TSQL.

Analytics & Visualization: PowerBI, Tableau, Kibana, Looker Studio, RAWGraphs, Office Suite (Advanced VBA).

Specialized AI: LLM Orchestration (MCP), Multi-agent Systems, Privacy-Preserving Analytics (HyperLogLog), NLP, Computer Vision, .

Builder Toolkit (Full-stack Legacy): Shell (Bash/Batch), Git, HTML5, CSS3, JavaScript, C/C++ (Arduino/IoT), Django, Flask, Guinicorn, Unraid OS.

COURSES

Microsoft Certified: Azure AI Fundamentals (AI-900), Azure Machine Learning, Computer Vision on Azure, Natural Language Processing on Azure.

Google Cloud: Digital Transformation with Google Cloud.

Specialized: The Complete Cyber Security Course (Station X), Profession AI NLP, Data Analyst to Data Scientist (Aspire Journeys).

Hands-on: Arduino from basic to advanced (FabLab Milano).

EXTRA

- Hobby Farmer at the Family's Agricultural Business.
- IoT Hobbyist developing custom Smart Devices.
- Licensed Sailor with skippering/crew experience.
- WSET Qualifications | Wine & Spirit Education Trust

WORK EXPERIENCE

Nielsen - Technical Owner | Senior AI/ML Data Scientist



 Lugano, Switzerland

April 2022 – Ongoing 

Technical Leadership: Lead and provide architectural direction for a team of 5 engineers. Responsible for code quality, mentoring, and end-to-end delivery of high-throughput pipelines.

- Cost Optimization & Scalability: Orchestrated the migration of massive data pipelines to Airflow on Kubernetes. Optimized Spark resource allocation and migrated backends from Pandas to Polars, resulting in an annual cloud infrastructure saving of ~€800,000 (40-60% monthly reduction).
- Strategic Revenue Protection: Designed and implemented a deterministic statistical pipeline for audience measurement in cookieless environments. This initiative successfully protected products representing 25% of the company's digital revenue following the deprecation of 3rd-party cookies.
- High-Throughput Engineering: Architected production systems processing 1.66 billion records per execution, maintaining strict daily SLAs through performance engineering and distributed computing.
- Generative AI (AppSense): Developed a Siamese Network architecture with BERT encoders and Contrastive Learning for URL-to-app name imputation, solving complex many-to-many attribution problems for the US Mobile Panel.
- Agentic Ecosystem: Built internal AI agents using the Model Context Protocol (MCP), allowing stakeholders to query complex product data via natural language, increasing data accessibility across the department.

DXC Technology - Data Scientist



 Milan, Italy

November 2020 - April 2022 

Enterprise Intelligent Automation: Built a multi-modal AI platform (NLP, Computer Vision, OCR) for a major energy client as part of a €7M strategic project.

Real-time Monitoring: Developed outlier detection models for naval fleet predictive maintenance and error forecasting for a global streaming service.

TK Soluzioni - Data Scientist



 Saronno, Italy

January 2020 - November 2020 

Application of statistical, mathematical, and machine learning methods to analyze and optimize monitoring software for a big company in the publishing industry.

*** Prior experience during university includes roles as a Data Analyst, Web & Software Developer, and several business internships.

EDUCATION

B.Sc. Quantitative Methods	M.Sc. Statistics
<u>Thesis: "PERFORMANCE ANALYSIS AND FOOTBALL COMPETITION OUTCOME PREDICTION"</u>	<u>Thesis: "Separate collection of waste, a classification problem: Forecasting models and applications"</u>
Catholic University of Milan  2014/2015 – 2016/2017	Bicocca University of Milan  2017/2018 – 2018/2019

KEY OPEN SOURCE & FULL-STACK PROJECTS

scomp-link (PyPI): An end-to-end ML toolkit. Implemented a custom Lazy Loading system that reduced library import time from 5.2s to 5ms (99% optimization), now supporting 24 CLI commands for automated ML lifecycles.	PyPI GitHub Technical Deep Dive
Celebrimbot: An autonomous, multi-agent AI coding assistant for JetBrains IDEs. Features an offline-first architecture with local LLM inference and cloud fallback, managing tasks through a Tolkien-themed agentic workflow	GitHub JetBrains Marketplace Project Article
CardinalityKit: A comprehensive Python library for Privacy-Preserving Analytics. Implements probabilistic sketching algorithms (HyperLogLog, HyperReal) to estimate unique elements without storing sensitive raw IDs.	GitHub Article: The Upside Down of Data Medium Article
Starfleet Engineering (MCP): Developed a guide and framework for building custom Model Context Protocol servers, creating a "Universal Translator" for LLM-to-tool communication.	Project Guide Medium Article
ProjectCompass: A containerized execution engine and analysis catalog integrated with Docker and GitHub Container Registry, automating analysis scheduling and visualization.	GitHub Container Registry

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